

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Some Applications of Trigonometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 One right-angled triangle

1. A car drives away from the foot of a tower 30 m high. What is the angle of elevation of the top of the tower from the car when it is $10\sqrt{3}$ m from the foot? (a) 30° (b) 45° (c) 90° (d) 60°

CBSE 2026, 1 mark

SOLUTION

$$\tan \theta = \frac{\text{height}}{\text{distance}} = \frac{30}{10\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$\tan 60^\circ = \sqrt{3}$$

Answer: (d)

2. A kite is flying 150 m above the ground. Its string is straight and makes an angle of 30° with the ground. How long is the string? CBSE 2025, 1 mark

SOLUTION

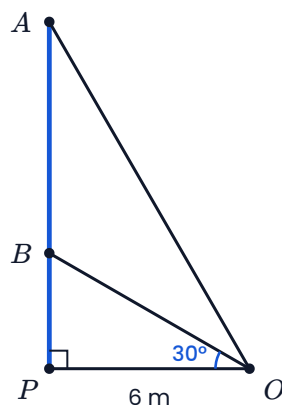
The height is the side opposite the 30° angle and the string is the hypotenuse.

$$\sin 30^\circ = \frac{150}{\text{string}}, \text{ so string} = \frac{150}{\frac{1}{2}}$$

Answer: 300 m

2 Two right-angled triangles

3. A radio tower PA is built in two sections: PB is the lower section and BA the upper one. Wires from a point O on the ground, 6 m from the foot P , hold the tower at B and at A . From O the angle of elevation of B is 30° and that of A is 60° . (i) Find the length of the wire OB . (ii) Find the length of the wire OA . (iii) Find AB , OR find the area of $\triangle OPB$. **CBSE 2026, 4 marks**



Angle $POB = 30^\circ$ and angle $POA = 60^\circ$

SOLUTION

(i) In $\triangle OPB$, $\cos 30^\circ = \frac{OP}{OB}$, so $OB = \frac{6}{\frac{\sqrt{3}}{2}} = \frac{12}{\sqrt{3}}$. $OB = 4\sqrt{3}$ m.

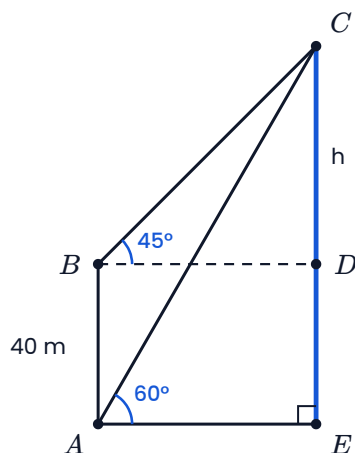
(ii) In $\triangle OPA$, $\cos 60^\circ = \frac{OP}{OA}$, so $OA = \frac{6}{\frac{1}{2}}$. $OA = 12$ m.

(iii) $PB = 6 \tan 30^\circ = \frac{6}{\sqrt{3}} = 2\sqrt{3}$ m and $PA = 6 \tan 60^\circ = 6\sqrt{3}$ m.

$AB = PA - PB = 6\sqrt{3} - 2\sqrt{3}$. $AB = 4\sqrt{3}$ m.

OR: area of $\triangle OPB = \frac{1}{2} \times OP \times PB = \frac{1}{2} \times 6 \times 2\sqrt{3}$. **Area = $6\sqrt{3}$ m².**

4. From a point A on the ground, the top C of a lighthouse CE is seen at an angle of elevation of 60° . From a deck at B , 40 m directly above A , the angle of elevation of C is 45° . D is the point on CE level with B , and $CD = h$ metres. (i) Write BD in terms of h . (ii) Write BC in terms of h . (iii) Find the height CE , taking $\sqrt{3} = 1.73$, OR find AE if $AC = 100$ m. **CBSE 2025, 4 marks**



SOLUTION

(i) In $\triangle BDC$, $\tan 45^\circ = \frac{CD}{BD}$, so $BD = CD$. $BD = h$ metres.

(ii) $\sin 45^\circ = \frac{CD}{BC}$, so $BC = \frac{h}{\frac{1}{\sqrt{2}}}$. $BC = \sqrt{2}h$ metres.

(iii) $AE = BD = h$ and $CE = h + 40$. In $\triangle AEC$, $\tan 60^\circ = \frac{h + 40}{h}$.

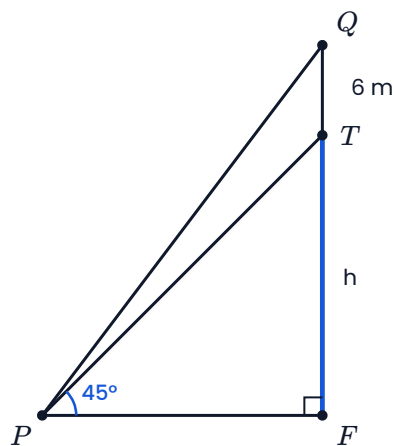
$$\sqrt{3}h = h + 40, \text{ so } h = \frac{40}{\sqrt{3} - 1} = \frac{40(\sqrt{3} + 1)}{2} = 20(\sqrt{3} + 1) = 20 \times 2.73 = 54.6$$

$CE = 54.6 + 40$. **The lighthouse is 94.6 m high.**

OR: in $\triangle AEC$, $\cos 60^\circ = \frac{AE}{AC}$, so $AE = 100 \times \frac{1}{2}$. $AE = 50$ m.

5. A 6 m pole stands on top of a tower. From a point P on the ground, the top of the pole is seen at an angle of elevation of 60° . From the top of the tower, P is seen at an angle of depression of 45° . Find the height of the tower and the distance of P from its foot. (Take $\sqrt{3} = 1.73$.)

CBSE 2024, 5 marks



FT is the tower and TQ is the pole

SOLUTION

Let the height of the tower be h m and the distance of P from its foot be d m.

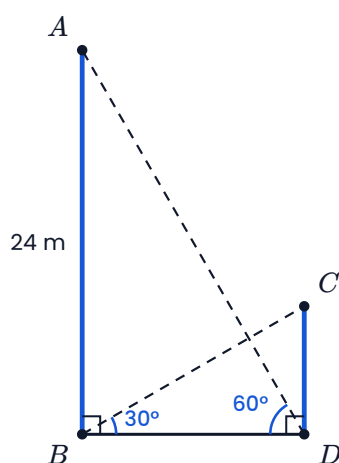
The angle of depression of P from the top of the tower is 45° , so the angle of elevation of the top of the tower from P is 45° : $\tan 45^\circ = \frac{h}{d}$, so $d = h$.

For the top of the pole: $\tan 60^\circ = \frac{h+6}{d}$, so $\sqrt{3}h = h+6$.

$$h = \frac{6}{\sqrt{3}-1} = \frac{6(\sqrt{3}+1)}{2} = 3(\sqrt{3}+1) = 3 \times 2.73 = 8.19$$

Answer: the tower is 8.19 m high and P is 8.19 m from its foot.

6. Two towers stand on level ground. The first is 24 m high, and from the foot of the second tower its top is seen at an angle of elevation of 60° . From the foot of the first tower, the top of the second is seen at an angle of elevation of 30° . Find the distance between the towers, the height of the second tower, and the length of a wire joining the two tops. **CBSE 2023, 5 marks**



SOLUTION

Let $AB = 24$ m be the first tower, $CD = h$ the second, and $BD = d$ the distance between them.

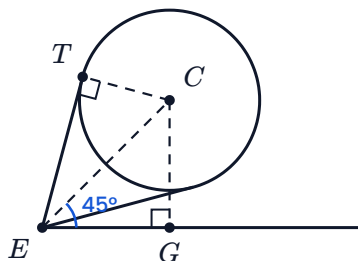
In $\triangle ABD$: $\tan 60^\circ = \frac{24}{d}$, so $d = \frac{24}{\sqrt{3}} = 8\sqrt{3}$ m ≈ 13.86 m.

In $\triangle CDB$: $\tan 30^\circ = \frac{h}{d}$, so $h = \frac{8\sqrt{3}}{\sqrt{3}} = 8$ m.

The tops differ in height by $24 - 8 = 16$ m. Wire $AC = \sqrt{(8\sqrt{3})^2 + 16^2} = \sqrt{192 + 256} = \sqrt{448} = 8\sqrt{7}$ m.

Answer: distance $8\sqrt{3}$ m; second tower 8 m; wire $8\sqrt{7}$ m ≈ 21.17 m

7. A spherical balloon of radius r subtends an angle of 60° at the eye of an observer, and the angle of elevation of its centre from the eye is 45° . Prove that the centre of the balloon is at a height of $\sqrt{2}r$. **CBSE 2023, 5 marks**



SOLUTION

Let E be the eye, C the centre of the balloon, ET a tangent to the balloon and CG the height of the centre above the level of the eye.

The two tangents from E make an angle of 60° , and EC bisects it, so $\angle CET = 30^\circ$.

$CT \perp ET$ (radius and tangent). In $\triangle CTE$: $\sin 30^\circ = \frac{r}{EC}$, so $EC = 2r$.

In $\triangle CGE$: $\sin 45^\circ = \frac{CG}{EC}$, so $CG = 2r \times \frac{1}{\sqrt{2}} = \sqrt{2}r$.

So the height of the centre is $\sqrt{2}$ times the radius. Proved.

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.